Continuous Distillation

Final Lab Report

Unit Operations Lab 5

April 6th, 2026

Group 3, Tuesday 9 AM

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1 Executive Summary

The objective of this experiment is to examine a two-component (binary) distillation column that separates ethanol and water operating at steady state through a continuous feed and determining how the location of the feed affected column efficiency. The experiment consisted of three subsections. First, the feed pump is calibrated by measuring volumetric flowrate against the ten positions of the dial on the console to achieve greater accuracy. The calibration graph produced a linear relationship with an R^2 value of 0.9884. The column is set to operate with a reflux ratio of 5:1, with the initial feed position set in the middle of the column. Liquid product samples will be taken from both the top and bottom to measure the ethanol composition. Overall column efficiency will be determined with a McCabe-Thiele analysis. Steady state was reached after around 25 minutes and 6 theoretical stages were determined with the McCabe-Thiele analysis, with an overall column efficiency of 66%. Finally, still under continuous operation, the effect of changing the feed position is determined. The top-feed position produced 3 theoretical stages and an overall efficiency of 33%. The middle-feed position provided better separation of the ethanol and showed that feed position affected column and separation efficiency.

2 Introduction

Distillation is one of the most used separation techniques used in chemical engineering. Used to produce very high purity products, it can be used in a multitude of feeds. Because of this it is used extensively in industry in chemical manufacturing, mainly in petroleum refinery (refining crude oil) and mixture purification [6]. With this experiment, ethanol-water distillation is relevant to show how different operating factors such as reflux ratio and feed position affect efficiency as these are important factors in producing high-quality products.

Classified as an equilibrium separation process as the gas and liquid phases are predicted to reach equilibrium at each of the contacting stages in the distillation column. As the concept of equilibrium at each stage is theoretical as this not possible in fact, chemical engineers have created models, such as column efficiency to analyze this process for the non-ideal behavior [2]. The objective of this lab is to examine the distillation at steady state of 10 wt% and 20 wt% mixture of ethanol and water and to investigate the effects of feed position with the column efficiency under continuous operation. Top and bottom product samples are collected to determine ethanol compositions, with McCabe Thiele analysis conducted to evaluate separation performance and estimate column efficiency. The effect of feed location is studied by comparing results obtained at different feed positions under the same operating conditions. The significance of this lab is shown as distillation columns don't achieve ideal equilibrium at all the stages, so experimental data is needed to compare the ideal stage predictions with actual column behavior.

The distillation columns used in this experiment are the Armfield UOP3 – Distillation Columns which is equipped with two interconnected units: a floor standing process unit and a bench-mounted control console with a PC connected to it [5]. The columns are self-contained making it suitable for practical lab work relevant in teaching the unit operations of separation and distillation [5].

3 Theory

Continuous distillation is one of the most widely used separation processes in chemical engineering. In this process, components (ethanol and water in this lab) are separated based on volatility within a column containing several trays where vapor-liquid equilibrium is achieved. Exchange between each equilibrium-stage (vapor flows to the top of the column while liquid moves downwards) combined with this volatility difference allows for a high degree of separation to be achieved, with the more volatile component being more abundant in the distillate (top) product and the less volatile component making up a greater part of the bottoms product.

Since the process is continuous, a feed is introduced to the column. In addition to this steady inflow of material, a portion of the vapor condensed in the reboiler is sent back to the column as reflux. The remainder is collected as the distillate product. A similar process happens at the bottom of the column with the liquid matriculating through the reboiler system. Since, at steady state, vapor-liquid equilibrium is achieved within each tray in the column, the positioning of the factors that affect the amount of vapor and liquid entering or exiting the column affect the overall equilibrium, and therefore the separation efficiency, of the column. For similar reasons, the same factors may affect the quality of the separation and the purity of the product leaving the column [1,2].

Specifically, in the case of continuous distillation, feed placement, condition, and feed rate are major considerations that impact column efficiency. Initially, the feed will enter near the middle of the column, as this is consistent with what is often observed in industrial applications of distillation, and it also allows for separate operating lines for the rectifying (above feed) and stripping (below feed) sections of the column to be constructed in a theoretically consistent and meaningful way. Additionally, the rate at which liquid returns through the column through the reflux drum (reflux ratio) must be considered. This ratio is defined as the ratio of liquid returning to the column vs. liquid being withdrawn as the distillate product. Higher reflux ratios generally imply greater separation of components but is usually also more energy intensive [2].

The efficiency and separation efficacy of a column can be determined from a mass balance. However, this method becomes convoluted when considered more complex systems with many components or partial reflux, which often require simulation to properly assess. In the case of binary distillation, McCabe-Thiele Analysis can be implemented to understand the impact of process variable changes on the system. Knowing the composition of the feed, top, and bottoms products, a McCabe-Thiele diagram can be constructed by plotting the rectifying operating line, stripping operating line, feed line (*Equations 2-4, Table 1)* and VLE data for the binary mixture. The number of theoretical stages can be found geometrically from the resulting figure [1]. This can be compared to the number of actual stages to obtain the overall column efficiency (*Equation 6*, *Table 1*). This information can be corroborated through the calculation of Murphee plate efficiency (*Equation 5*, *Table 1*), which more explicitly considers mechanical inefficiency throughout the system. Obtaining these two values for different test conditions allows for a well-rounded analysis of the separation and energy efficiency of a specific case, which is valuable when there are many process variables, as in the case of continuous distillation [3]. Examination of non-steady state column operation and the development of the temperature profile of the column under different conditions can also yield useful information about optimal operation.

Table 1 : Equations

Title	Equation	Components
1. Component Mass Balance		F = feed flowrate D = distillate flowrate B = bottoms flowrate = molar feed composition = molar distillate composition = molar bottoms composition
2. Rectifying Operating Line		R = reflux ratio
3. Stripping Operating Line		L'= liquid flow between trays V' = vapor flow between trays
4. Feed Line (q-line)		q = feed condition
5. Murphee Plate Efficiency		= equilibrium data

6. Overall Column Efficiency		

4 Results

<u>Experiment A: Feed Pump Calibration</u>

The feeding tube was disconnected from the column and placed into a graduated cylinder while the pump was operated at multiple dial settings. For each of the 10 settings, the volume collected was measured, and the time was recorded using a stopwatch. The volumetric flow rate (mL/min) was calculated for each setting, and a calibration curve shown below in *Figure 1* was generated to determine feed flow rates during experiment B and C.

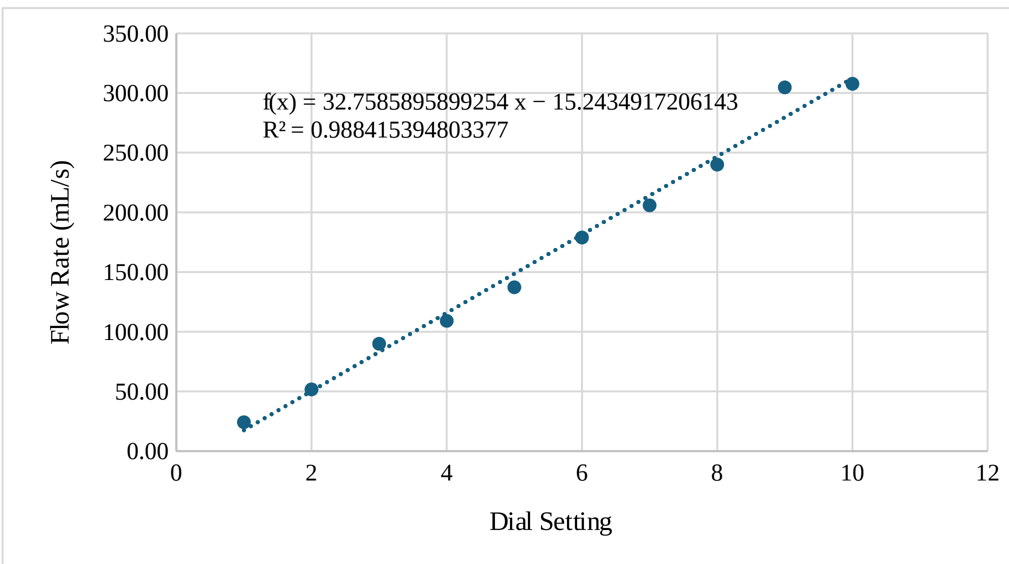

Figure 1: Calibration Curve for Pump Speed

Figure Figure 1 indicates that the flowrate increases linearly with increasing dial setting, showing a strong correlation to the best fit line equation. Deviations were small but can be attributed to improper collection methods, tubing deformation, and possibly pump inefficiencies.

<u>Experiment B: Steady State Distillation at Middle Feed</u>

Tray temperatures 1-9 (T1-T9) were monitored using the computer connected to the Armfield console and apparatus. Once steady state was established, the reflux ratio was set to 5:1, the feed was introduced at the midpoint of the column, and bottoms flow was initiated. After ~400 mL of feed was processed, distillate and bottoms samples were collected from the overhead valve and reboiler outlet respectively. A total of three sets of samples were taken at 5-minute intervals and at each sampling point temperature at each tray (T1-T9) was recorded, and ethanol composition of each sample was measured using the densitometer (SNAP 41). As shown in the table below, temperature was monitored in trays 1-9 which indicates that steady state was reached after ~25 minutes. The consequent samples of distillate and bottoms product were collected at ~30 minutes and finally ~35 minutes.

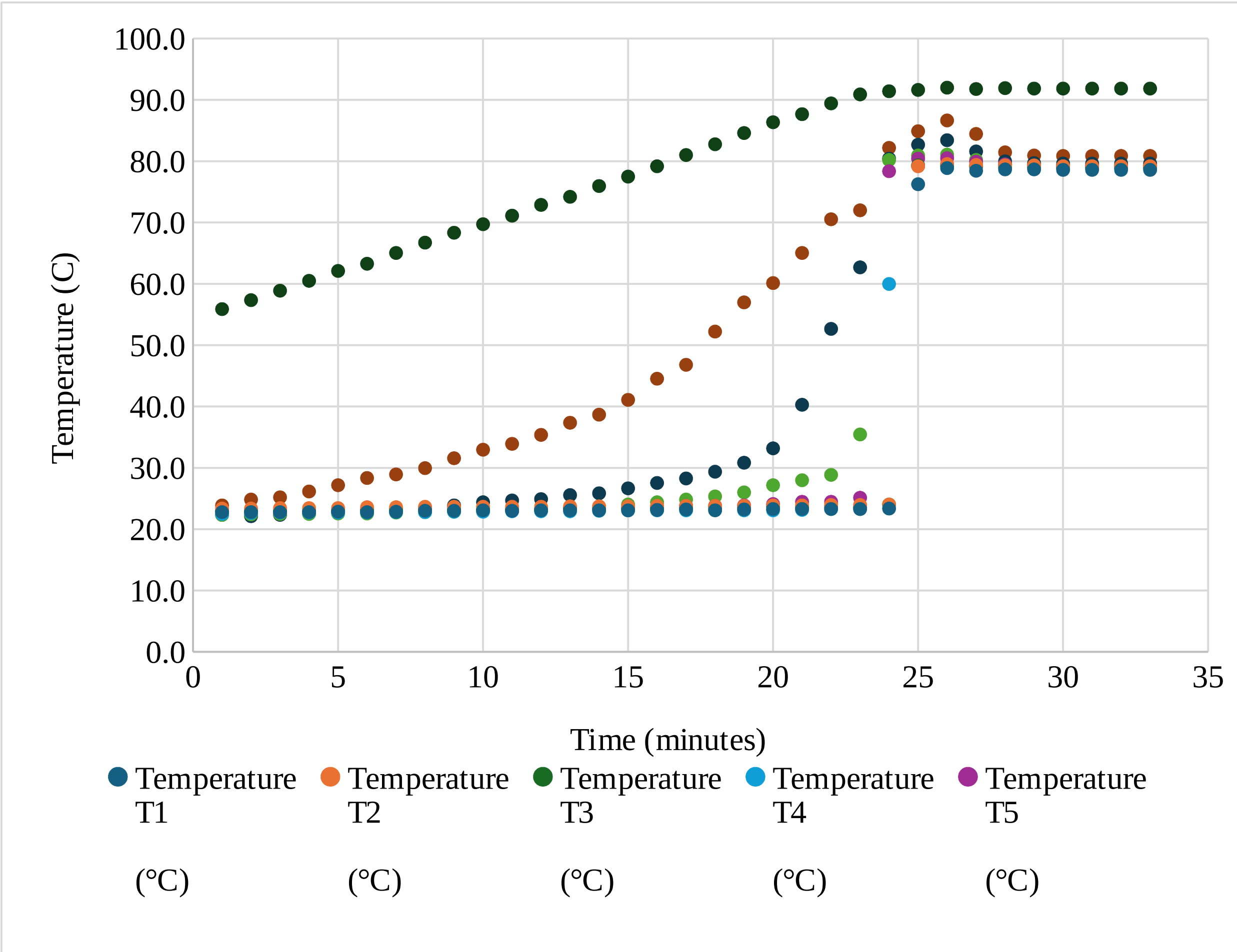

Figure 2: Temperature Profiles of Column Trays (T1–T9) vs. Time for Midpoint Feed

After ~400 mL of feed was processed, distillate and bottoms samples were collected from the overhead valve and reboiler outlet respectively. A total of three sets of samples were taken at 5-minute intervals and at each sampling point temperature at each tray (T1-T9) was recorded, and ethanol composition of each sample was measured using the densitometer (SNAP 41).

Next, a McCabe-Thiele analysis was conducted using the experimentally measured distillate and bottoms compositions along with the imposed reflux ratio of 5:1. The feed was assumed to be a saturated liquid, resulting in a vertical q-line at the feed composition. The equilibrium data for the ethanol-water system was used to construct the equilibrium curve, and operating lines were generated based on mass balances around the column. The resulting diagram was used to estimate the number of theoretical stages required for the observed separation and to compare ideal behavior with experimental performance.

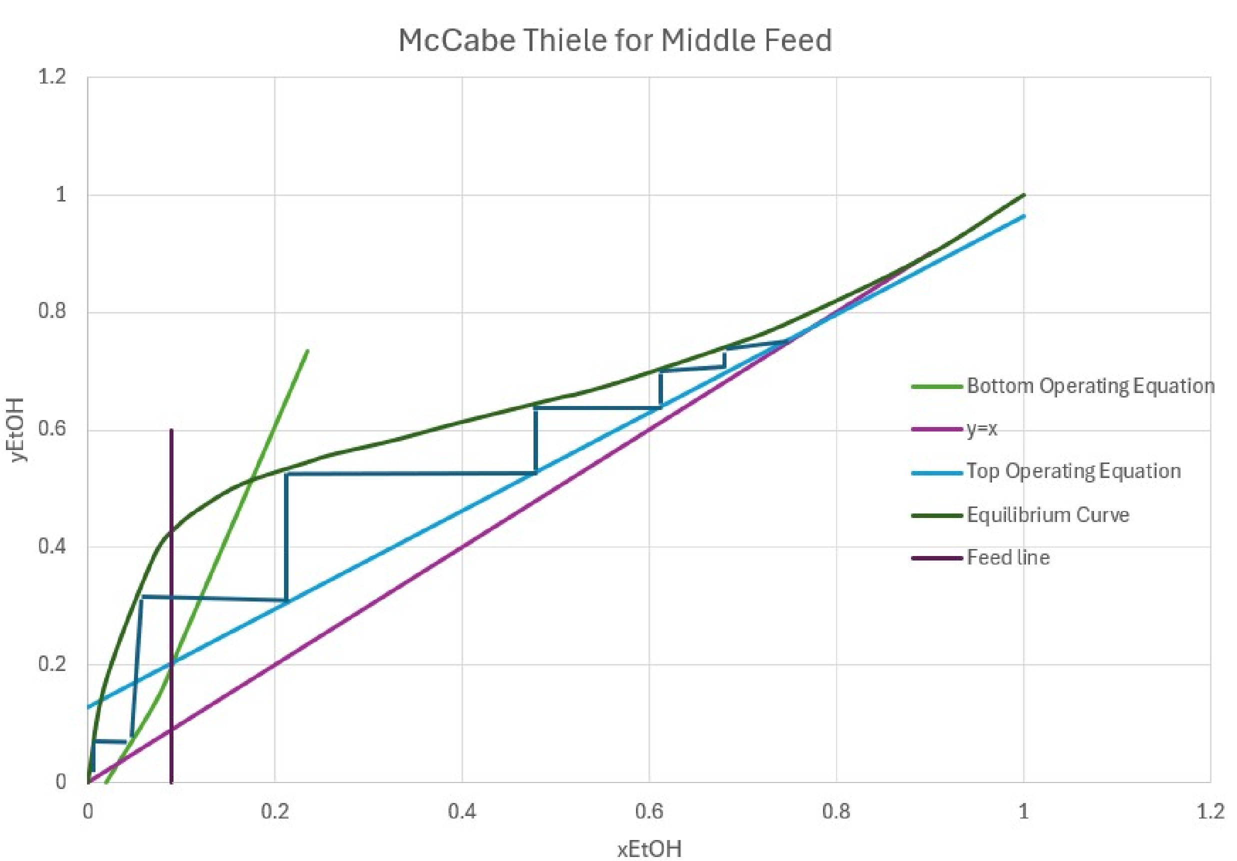

Figure 3: McCabe–Thiele Diagram for Ethanol–Water Separation at Midpoint Feed

Last, the column efficiency is calculated, assuming that the actual number of stages is 8+1, where 1 is added due to the reboiler. From the table above, it is noted that there are 6 theoretical stages making the efficiency 66%.

Experiment C: Steady State Distillation at Top Feed

For this experiment, the feed location was moved from the midpoint to above the trop tray, and the column was brought again to a steady state under total reflux by monitoring tray temperatures. Similar setup procedures to experiment B were followed and the reflux ratio was set to 5:1. Top and bottom product samples were collected and repeated three times at 5-minutee intervals.

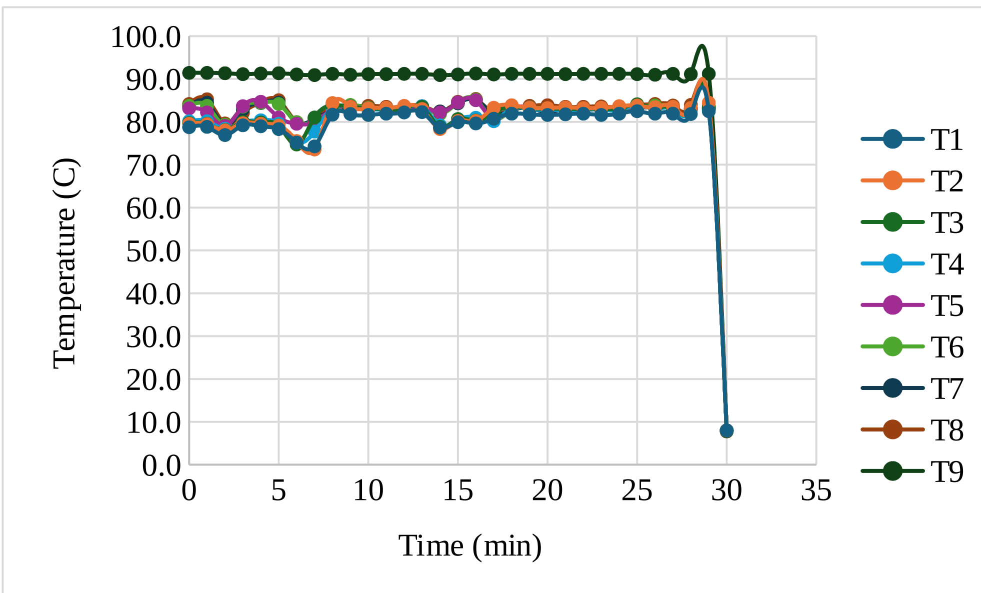

Figure 4: Temperature Profiles of Column Trays (T1–T9) vs. Time for Top Feed

Comparably to experiment B, a McCabe Thiele analysis is performed, and the theoretical number of stages is calculated. This is shown below in *Figure 10*. The resulting diagram will again aid in estimating the number of stages as well as the comparison of ideal behavior with experimental performance.

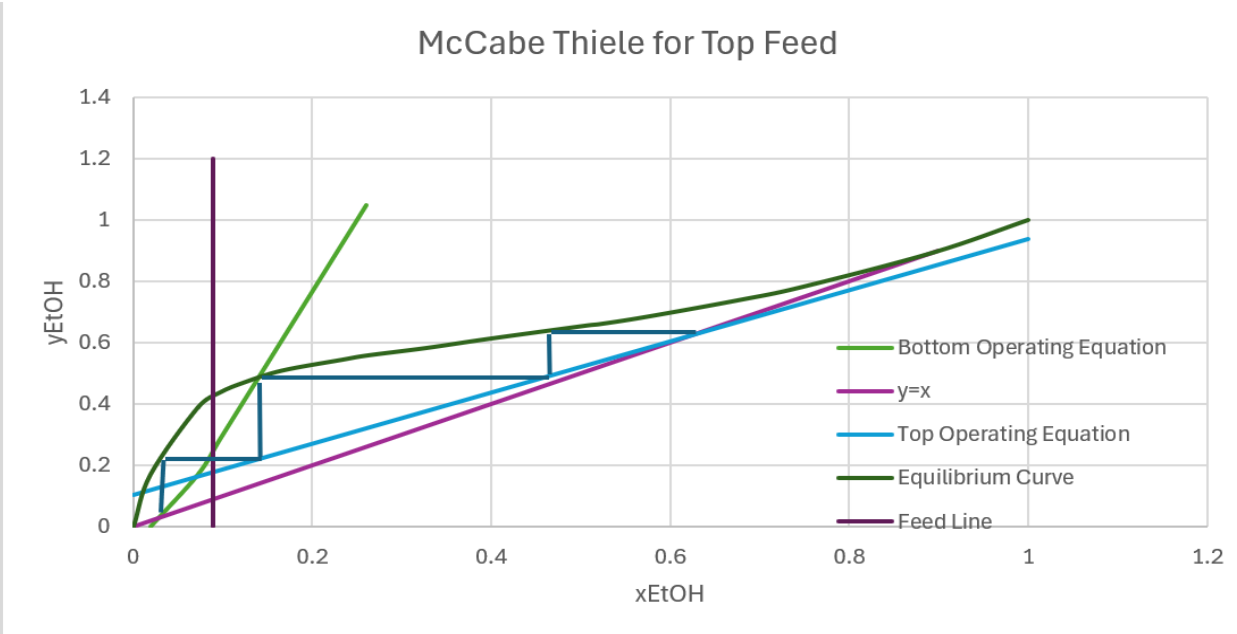

Figure 5: McCabe–Thiele Diagram for Ethanol–Water Separation at Top Feed

Similarly, the column efficiency is calculated assuming 9 actual stages and from the graph we can see the number of theoretical stages is 3. The efficiency is 33%.

5 Discussion

The results from this experiment demonstrate the fundamental behavior of a continuous distillation column and highlight the impact of operating conditions, specifically feed location, on separation performance.

In Experiment A, the calibration curve (Figure 1) showed strong linear relationship between pump dial setting and volumetric flow rate, with an R^2 value of 0.9884. This confirms that that the peristaltic pump operates predictably across the tested range, allowing accurate control of feed flow rate in subsequent experiments. Minor deviations from linearity can be attributed to experimental limitations such as manual timing errors, slight variation in tubing elasticity, and pulsation inherent to pumps. Nevertheless, the calibration was sufficiently accurate to ensure reliable flow rate selection during steady-state operation.

For Experiments B and C, the temperature profiles across trays provide insight into the dynamic behavior and approach steady state. As shown in the temperature vs. time data, all trays exhibit gradual increase in temperature followed by stabilization after approximately 25 minutes. This behavior is consistent with the expected startup of a distillation column, here heat input from the reboiler drives vapor upward and establishes vapor-liquid contact across trays. The temperature gradient observed along the column includes higher temperatures near the reboiler (T9) and lower temperatures near the condenser (T1). This confirms proper separation, as more volatile ethanol concentrates towards the top while water-rich liquid remains at the bottom. This aligns with distillation theory, which assumes countercurrent contact and equilibrium stages [2].

The McCabe-Thiele analysis provides a theory to evaluate column performance. For the middle feed configuration in Experiment B, approximately 6 theoretical stages were determined, corresponding to an overall efficiency of 66% when compared to the actual number of stages, 9. In contrast, the top feed configuration in Experiment C yielded to only 3 stages, resulting in a significantly lower efficiency of 33%. This greatly decreases the overall efficiency and highlights the importance of feed location in distillation design. When the feed is introduced near the midpoint, both rectifying and stripping sections are effectively utilized, maximizing mass transfer and separation efficiency. However, when the feed is introduced at the top, the stripping section dominates while the rectifying section becomes underutilized, leading to poorer separation.

Although the McCabe-Thiele method assumes ideal vapor-liquid equilibrium on each stage, the experimental results clearly indicate deviations from this assumption. In practice, trays do not achieve complete equilibrium due to finite mass transfer rates, imperfect mixing, and hydraulic limitations. This explains why the number of theoretical stages is lower than the actual number of trays and why efficiencies are less than 100%. The concept of Murphree efficiency accounts for this discrepancy and reflects the non-ideal nature of real distillation systems [2]. Additionally, heat losses to the surroundings, especially in a laboratory-scale column, further reduce efficiency by disturbing the assumed constant molar overflow.

Several sources of experimental error may have influenced the results. Temperature measurements may have slight inaccuracies due to sensor lag or calibration drift. Composition measurements using the densitometer could introduce uncertainty, particularly if samples were not fully equilibrated or contained temperature variations. Flow rate fluctuations from the pump and slight inconsistencies in maintaining the reflux ratio could also impact the operating lines used in the McCabe-Thiele analysis. Furthermore, the assumption that the feed is a saturated liquid may not have been perfectly met, introducing additional deviation from theoretical predictions.

Overall, the results demonstrate good qualitative agreement with distillation theory. The expected temperature gradients, steady-state behavior, and dependence of efficiency on feed location were all observed. The experiment confirms that optimal feed placement is critical for maximizing separation efficiency and that real systems deviate from ideal models due to non-equilibrium effects and energy losses. Future improvements could include better insulation to minimize heat loss, more precise flow control, and repeated trials to reduce experimental uncertainty.

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6 Appendix

6.1 Appendix I: Data Tabulation/Graphs

Table 2: Feed Pump Calibration Data

Dial Setting	Flow Rate (mL/min)
1	24.18

2	51.68
3	89.91
4	109.09
5	137.24
6	178.93
7	205.88
8	240.00
9	304.69
10	307.69

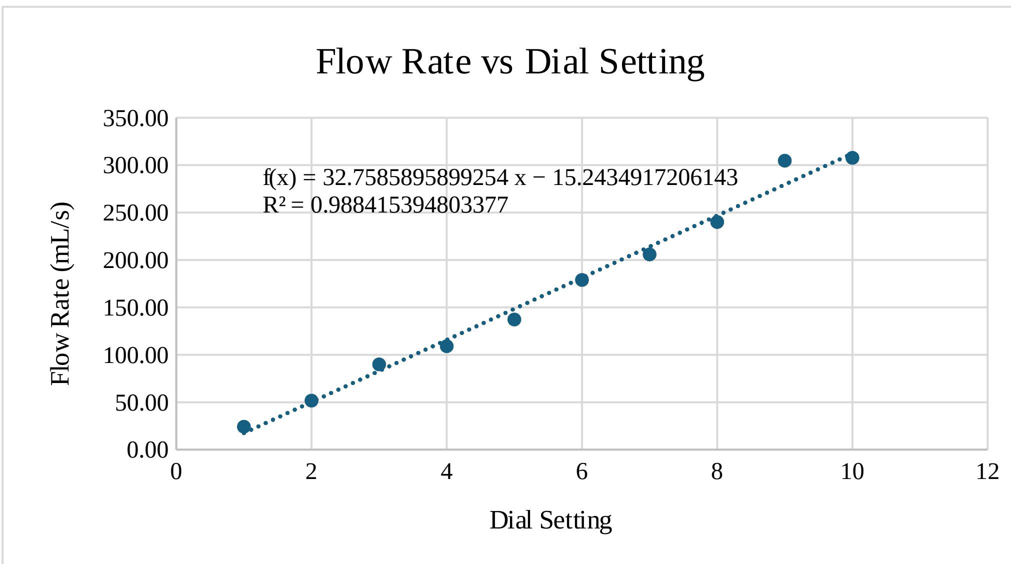

Figure 6: *Calibration Curve for Pump Speed*

Table 3: Experiment B Data

Tria l	Top Product (% v/v EtOH)	Bottoms Product (% v/v EtOH)	x_D	x_B	Ethanol (mol, top)	Water (mol, top)	Ethanol (mol, bottoms)	Water (mol, bottoms)	Total Moles (bottoms)
1	91.49	14.05	0.7684	0.04803	1.5687	0.8204	0.2406	4.7697	5.0103
2	91.93	12.72	0.7763	0.04754	1.5744	0.8222	0.2178	4.8355	5.0533
3	92.08	12.43	0.7802	0.04744	1.5769	0.8242	0.2129	4.8472	5.0601

Table 4: Experiment B Temperature Profile (Steady State)

Tray	Temperature (°C)
T1	78.6
T2	79.2
T3	79.0
T4	78.9
T5	79.2
T6	79.2
T7	79.6
T8	80.9
T9	91.8

Table 5: Experiment C Data

Trial	Top Product (% v/v EtOH)	Bottoms Product (% v/v EtOH)	x_D	x_B	Ethanol (mol, top)	Water (mol, top)	Ethanol (mol, bottoms)	Water (mol, bottoms)	Total Moles (bottoms)
1	85.14	13.87	0.638 8	0.04734	1.4581	0.8246	0.2375	4.7797	5.0172
2	83.22	13.16	0.604 8	0.04468	1.4252	0.9312	0.2254	4.8191	5.0445

3	82.61	13.01	0.5945	0.04412	1.4148	0.9650	0.2228	4.8274	5.0502

Table 6: Experiment C Temperature Profile (Steady State)	
Tray	**Temperature (°C)**
T1	81.9
T2	83.3
T3	83.0
T4	82.8
T5	83.0
T6	83.1
T7	83.1
T8	83.5
T9	91.1

Table 7: Column Temperature Data											
Sample	**Time**	**Elapsed (min)**	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**	**T7**	**T8**	**T9**
26	09:43	25	78.9	79.5	79.3	79.4	80.5	81.1	83.4	86.6	92.0
27	09:44	26	78.4	79.4	79.1	79.2	79.9	80.2	81.6	84.4	91.8
28	09:45	27	78.7	79.3	79.2	79.0	79.5	79.5	80.0	81.4	91.9
29	09:46	28	78.7	79.3	79.0	79.0	79.2	79.2	79.7	80.9	91.8
30	09:47	29	78.6	79.2	79.0	78.9	79.2	79.2	79.6	80.9	91.8
31	09:48	30	78.6	79.2	79.0	78.9	79.2	79.2	79.	80.9	91.8

									6		
32	09:49	31	78.6	79.2	79.0	78.9	79.2	79.2	79.6	80.9	91.8
33	09:50	32	78.6	79.2	79.0	78.9	79.2	79.2	79.6	80.9	91.8
34	09:51	33	78.6	79.1	78.9	78.7	79.0	79.0	79.5	80.3	91.7
35	09:52	34	78.5	79.0	78.7	78.7	79.0	79.0	79.3	80.7	91.6
36	09:53	35	78.4	79.0	78.7	78.6	79.0	79.0	79.4	80.3	91.6
37	09:54	36	78.5	79.0	78.8	78.6	79.0	79.1	79.5	80.4	91.6
38	09:55	37	78.6	79.0	78.7	78.6	79.0	79.1	79.4	80.4	91.6
39	09:56	38	78.4	79.0	78.7	78.6	79.0	79.0	79.3	80.6	91.6
40	09:57	39	78.5	79.0	78.7	78.6	79.0	79.0	79.4	80.5	91.6
41	09:58	40	78.5	79.0	78.7	78.7	79.0	79.2	79.5	80.6	91.8
42	09:59	41	78.3	79.0	78.7	78.7	79.2	79.7	79.9	81.0	91.6
43	10:00	42	78.4	79.0	78.7	78.7	79.2	79.5	79.8	81.1	91.6
44	10:01	43	78.4	79.0	78.8	79.0	81.6	82.1	81.7	82.0	91.6
45	10:02	44	78.3	78.9	78.7	79.0	83.4	83.5	83.3	83.2	91.3
46	10:03	45	78.6	79.1	78.9	79.2	82.9	83.2	83.1	83.8	91.4
47	10:04	46	78.4	79.2	79.2	79.6	82.5	83.3	81.	84.1	91.4

									1		
48	10:05	47	78.6	79.2	79.1	79.5	82.2	82.9	83.1	83.6	91.3
49	10:06	48	78.7	79.3	79.2	79.6	82.6	83.1	83.1	83.6	91.5
50	10:07	49	78.6	79.2	79.1	79.5	82.3	83.1	83.1	83.7	91.4
51	10:08	50	78.7	79.2	79.1	79.6	82.6	83.0	83.1	83.6	91.3
52	10:09	51	78.7	79.2	79.2	79.7	82.4	82.7	83.0	83.5	91.5
53	10:10	52	78.6	79.2	79.2	79.7	82.0	83.0	82.8	83.5	91.3
54	10:11	53	78.6	79.2	79.2	79.6	82.4	83.1	83.1	83.6	91.5
55	10:12	54	78.6	79.3	79.2	79.6	82.1	83.1	82.9	83.3	91.5
56	10:13	55	78.6	79.2	79.2	79.6	82.1	82.8	83.0	83.3	91.4
57	10:14	56	78.6	79.2	79.3	79.5	82.2	82.9	82.8	83.4	91.3
58	10:15	57	78.7	79.2	79.2	79.5	82.2	83.0	83.1	83.7	91.4
59	10:16	58	78.7	79.4	79.6	80.3	83.7	84.0	83.9	84.3	91.3
60	10:17	59	78.4	79.4	79.4	79.9	82.5	83.5	83.7	84.1	91.3

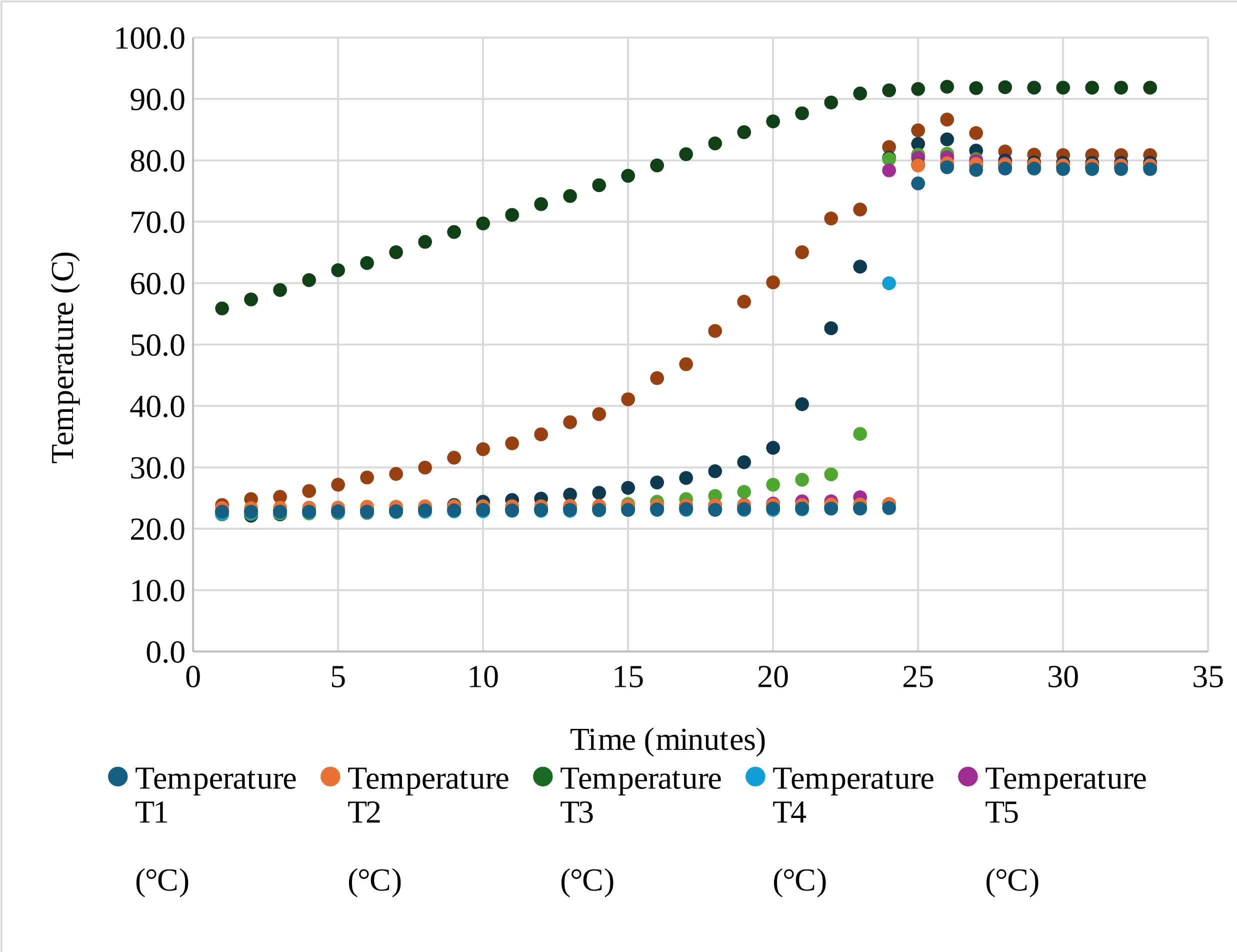

Figure 7: Temperature Profiles of Column Trays (T1–T9) vs. Time for Midpoint Feed

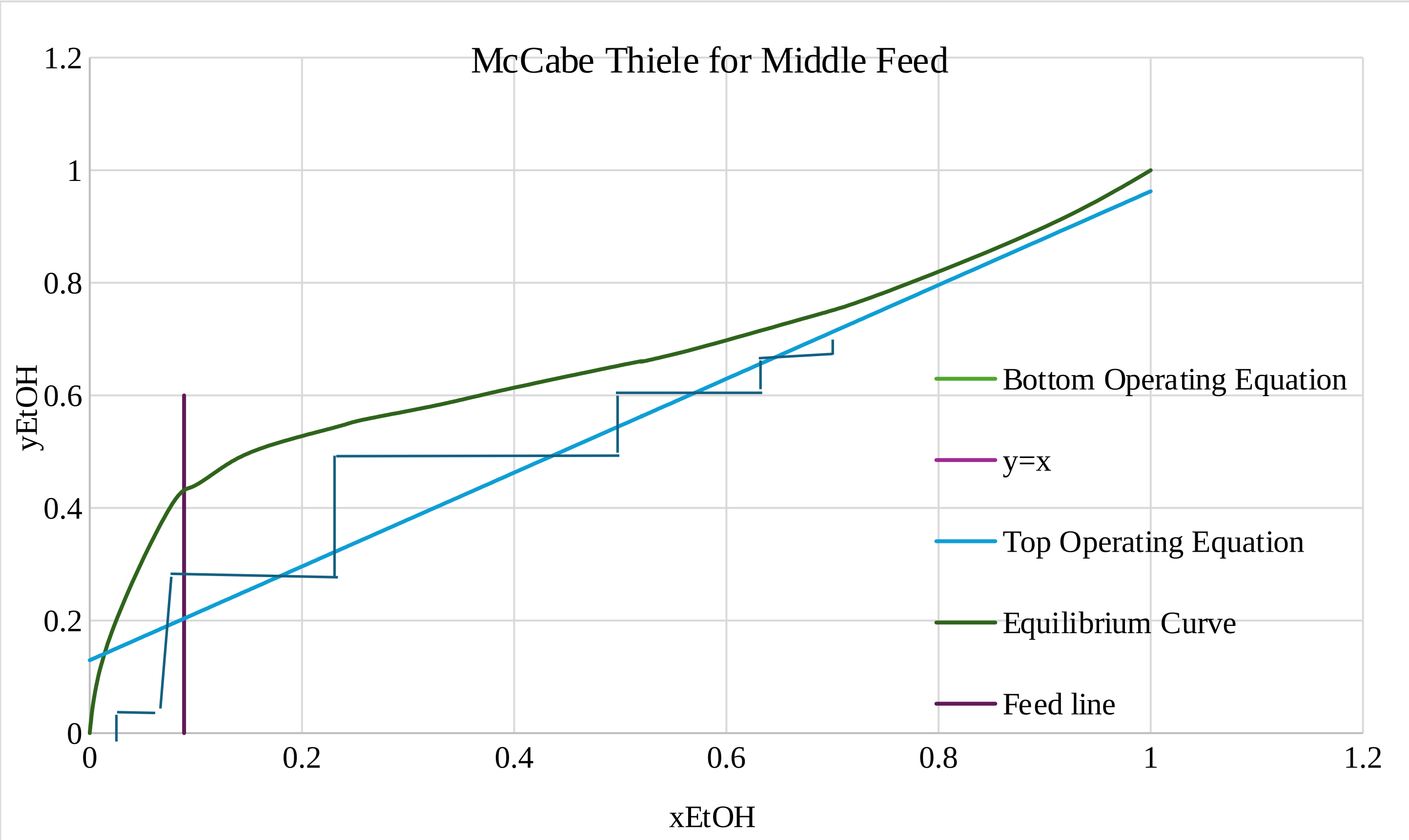

Figure 8: McCabe–Thiele Diagram for Ethanol–Water Separation at Midpoint Feed

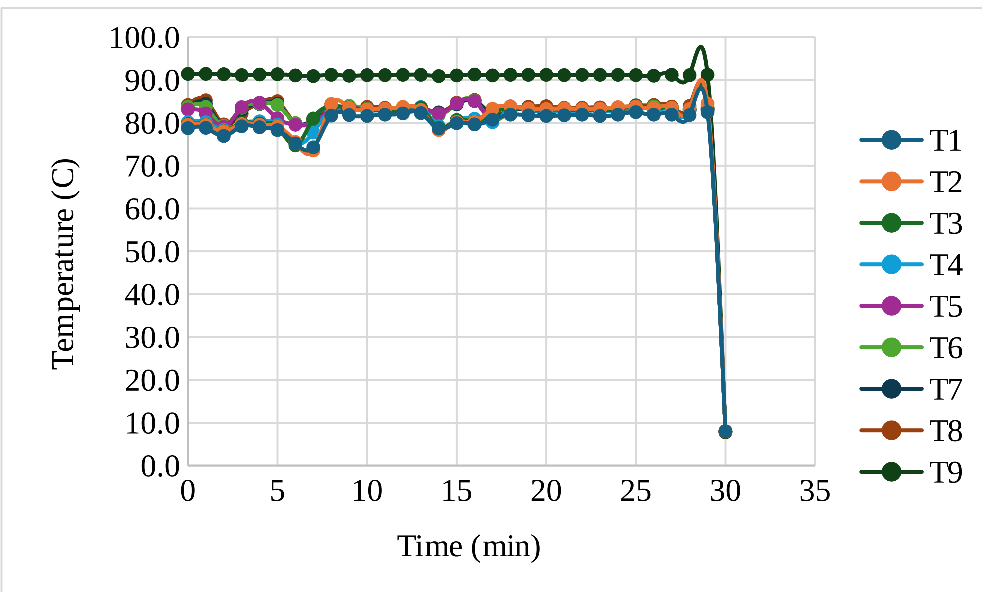

Figure 9: Temperature Profiles of Column Trays (T1–T9) vs. Time for Top Feed

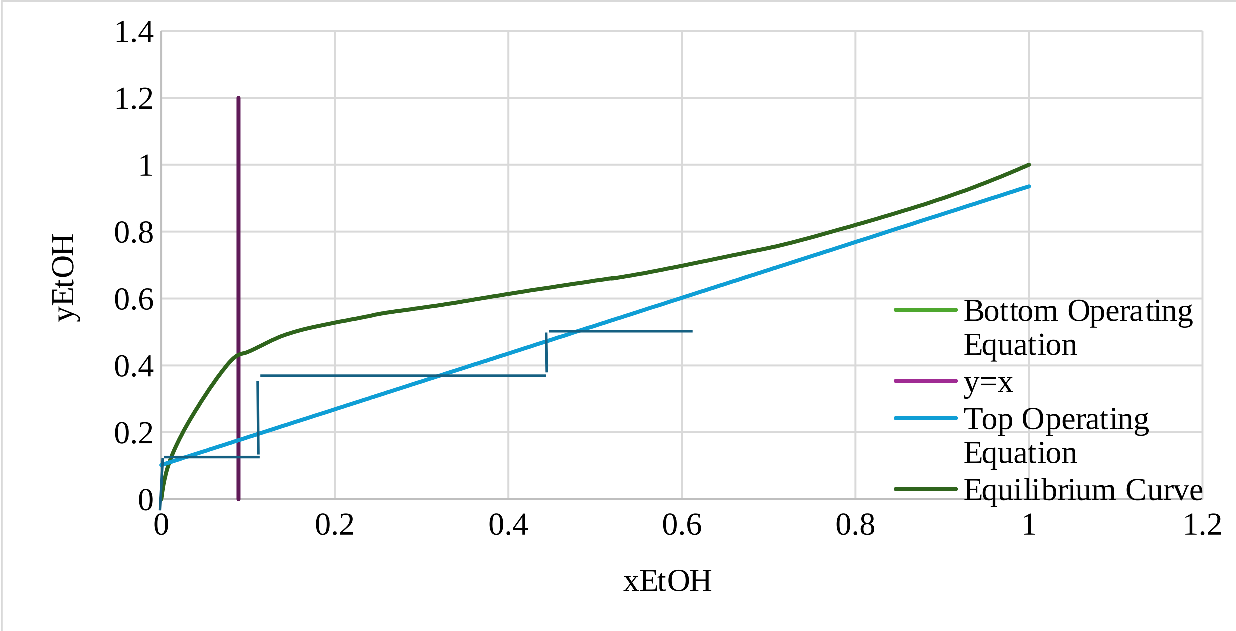

Figure 10: McCabe–Thiele Diagram for Ethanol–Water Separation at Top Feed

6.2 Appendix II: Error Analysis

All measurements in this experiment contain both random and systematic error, which affect the precision and reliability of the results. A quantitative analysis of error was performed using the repeated product composition measurements collected during each experiment.

Table 8: Summary of Quantitative Error Analysis

Experiment	**Stream**	**Mean**	**Std. Dev.**	**RSD (%)**	**95% CI (±)**
B	Distillate	0.7763	0.00708	0.91%	0.0176
B	Bottoms	0.04767	0.000314	0.66%	0.00078
C	Distillate	0.6127	0.02315	3.78%	0.0575
C	Bottoms	0.04538	0.001724	3.80%	0.00428

As shown in the table, Experiment B produced more precise results, with low RSD values (less than 1%) for both streams. In contrast, Experiment C showed significantly higher variability, especially in the distillate, which had the largest uncertainty (±0.0575). This indicates that the system was less stable when the feed position was changed.

The bottoms compositions in both experiments were more consistent than the distillate values, suggesting that the overhead product is more sensitive to changes in operating conditions such as reflux ratio, heat input, and feed location.

The main source of random errors were small fluctuations in column operation, including variations in feed flowrate, reflux ratio, and temperature. These fluctuations contribute to the spread in repeated measurements, which is reflected in the standard deviation and RSD values in the table.

Several systematic errors were also present. First, the McCabe-Thiele analysis assumes ideal stage equilibrium and no heat loss, but in practice the trays are not 100% efficient and heat is lost from the column. This leads to deviations between theoretical and experimental results. Second, measurement uncertainty from the densitometer likely contributed to error. Since samples were taken at elevated temperatures, any inconsistency in temperature control would affect density readings and therefore composition calculations. Finally, incomplete steady state operation may have affected the results. If the column had not fully stabilized before sampling, the measured compositions would not represent true steady-state conditions. This is consistent with the higher variability observed in Experiment C.

Overall, the data in the table shows that the experiment produced reasonable precision, particularly for Experiment B. However, the increased uncertainty in Experiment C suggests that changes in feed position made the system more sensitive to disturbances. While these errors introduce some uncertainty, the results still clearly demonstrate the expected separation behavior of the distillation column.

6.3 Appendix III: Sample Calculations

<u>DATA ANALYSIS</u>

1. **Experiment B Trial 1**

   Flow Rate

   The sample flow rate was calculated from:

   For the top product:

   For the bottom product:

2. **Converting alcohol concentration to mole fraction of ethanol in the distillate,**

   A basis of 100 mL of liquid sample was used.

   For the top sample:

   - Ethanol volume = 91.49 mL
   - Water volume = mL

   Ethanol mass

Water mass

Ethanol moles

Water moles

Total moles

Mole fraction of ethanol in distillate

3. Average product compositions for Experiment B

Three steady-state samples were taken for Experiment B. The average compositions used in the analysis were calculated as follows:

4. Feed composition used in the McCabe-Thiele analysis

The feed was prepared as 20 wt% ethanol and 80 wt% water. To convert this to mole fraction, a 100 g basis was used.

Ethanol moles

Water moles

Feed mole fraction

ERROR ANALYSIS

1. **Experiment B: Distillate composition,**

   Replicate values:

   Mean

   Range

   Standard deviation

Relative standard deviation

95% confidence interval

6.3 Appendix IV: Individual Team Contributions

Table 3: Individual Team Contributions	
Contribution	**Team Member**
Executive Summary, Introduction	Noah Guale
Theory	Basil Hashim
Results	Jazmin Aca
Discussion	Hiba-Batool Naqvi
Appendix I, II, III	Angelica Ramirez
References	All